

SSY156 Modeling and control of mechatronic systems

Lab report 1

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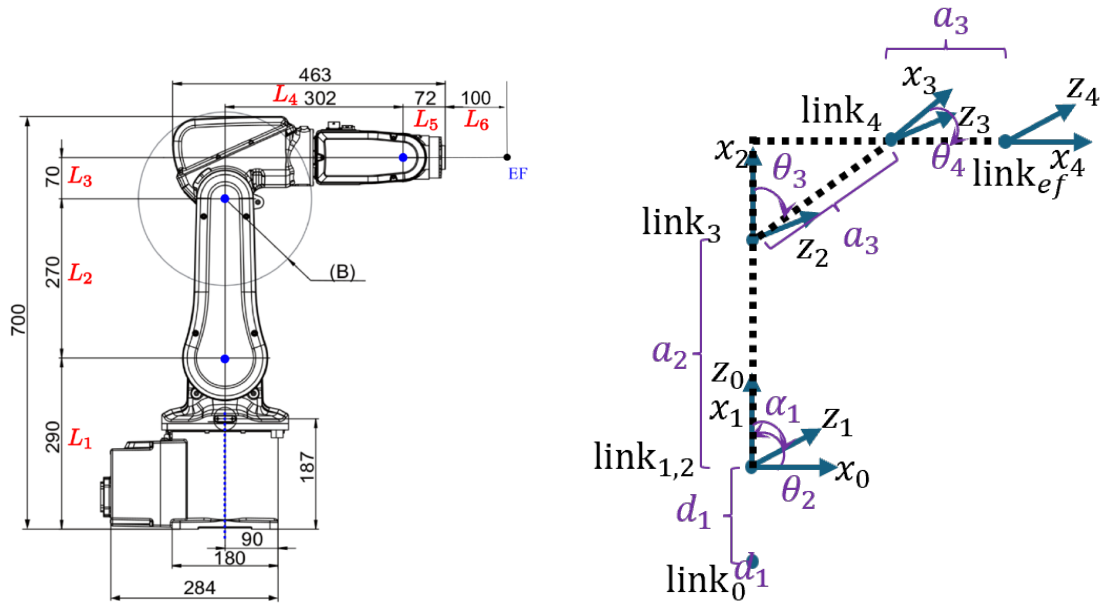
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Introduction

This lab report details the development and validation of kinematic and differential kinematic models for the 4-DOF ABB IRB 120 and the 7-DOF KUKA IIWA 7 robots. We derive the forward kinematics using Denavit-Hartenberg (DH) parameters, compute the analytical Jacobian matrices, and validate them against numerical velocity derivatives via MATLAB and Simulink simulations. Finally, we analyze the robots' manipulability ellipsoids and singular configurations to evaluate their end-effector motion capabilities.

Exercise 1

- (a) Figure 1 illustrates the coordinate frames assigned to the ABB IRB 120 robot based on its physical model.



(a) ABB robot physical model

(b) ABB robot coordinate frames

Figure 1: Illustration of the drawn reference frames for the manipulator.

- (b) Following the assigned frames and the Denavit-Hartenberg (DH) convention, the resulting DH parameters for the ABB IRB 120 robot are summarized in Table 1.

Link	θ_i	d_i	α_i	a_i
1	q_1	L_1	$-\pi/2$	0
2	$q_2 - \pi/2$	0	0	L_2
3	$q_3 + \arctan(\frac{L_4}{L_3})$	0	0	$\sqrt{L_3^2 + L_4^2}$
4	$q_4 + \arctan(\frac{L_3}{L_4})$	0	0	$L_5 + L_6$

Table 1: DH parameters for ABB IRB 120

Exercise 3

- (a) As demonstrated in Figure 2, the kinematic model is successfully validated through the 3D visualizer. For randomly generated joint configurations, the origins of the local coordinate frames (CFs) perfectly coincide with the physical joint locations of the robot model. This visual alignment confirms the accuracy of the derived Denavit-Hartenberg parameters and the resulting absolute Homogeneous Transformation matrices.

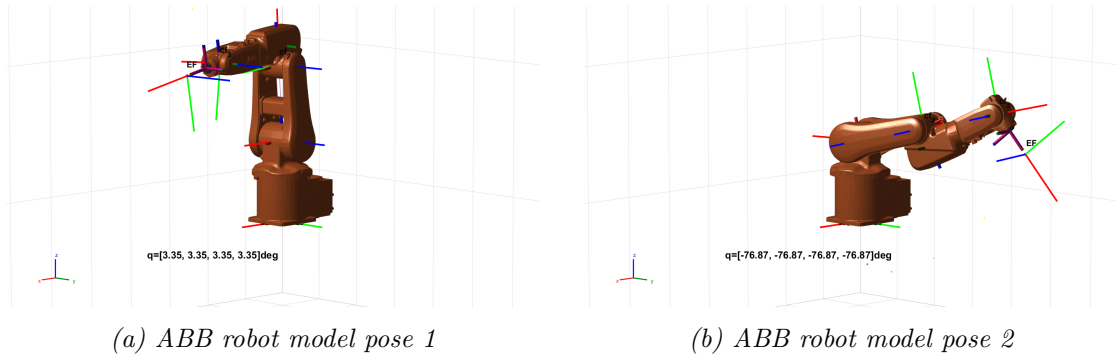


Figure 2: Visual validation of the kinematic frames under arbitrary joint configurations.

Exercise 5

- (a) The analytical Jacobian matrices derived for the ABB IRB 120 were rigorously and successfully validated against the numerical velocity derivatives using MATLAB and Simulink simulations. The differences between the analytical and numerical velocities were negligible (on the order of 10^{-3}), falling well within acceptable tolerances.

- (b) Figure 3 depicts the manipulability ellipsoid for the ABB IRB 120 at a specific joint configuration. The ellipsoid visually represents the end-effector's motion capabilities, where its principal axes and overall shape indicate the directions of maximum and minimum manipulability.

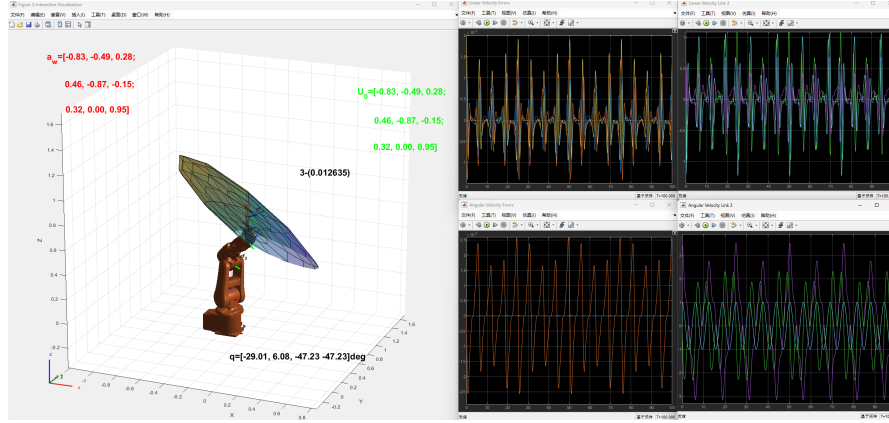
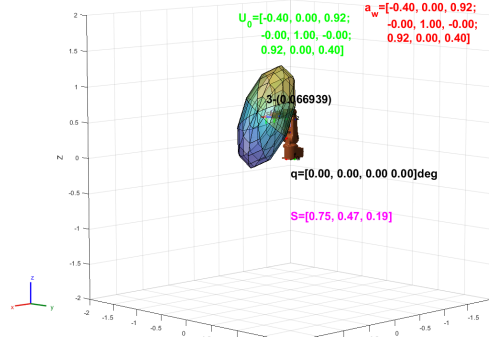


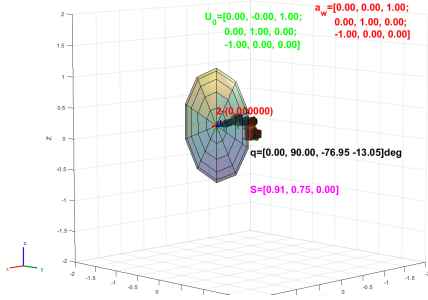
Figure 3: Manipulability ellipsoid for the robot at the specified joint configuration.

Exercise 6 and 7

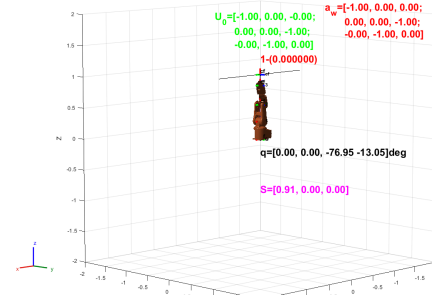
- (a) The shape of the manipulability ellipsoid is defined by the singular values of the Jacobian matrix. As the robot approaches a singular configuration, certain singular values approach zero, causing the ellipsoid to shrink or flatten along specific axes. This visually indicates a loss of manipulability in those directions. By observing the ellipsoid's transition from a full 3D volume to a flat 2D plane, and finally to a 1D line, we can clearly illustrate the loss of one or two degrees of freedom (DOFs).
- (b) In the 2-DOF case, the manipulator loses one degree of freedom, causing the manipulability ellipsoid to degenerate into a flat plane. This indicates that the end-effector's instantaneous velocity is restricted entirely to that plane. Here, we selected the joint configuration $q = [0, \pi/2, -\arctan(\frac{L_4}{L_3}), -\arctan(\frac{L_3}{L_4})]^T$. Conversely, in the 1-DOF case, the arm loses two degrees of freedom, and the ellipsoid further degenerates into a 1D line. This implies the end-effector can only translate along a single axis. For this extreme singularity, we chose the joint configuration $q = [0, 0, -\arctan(\frac{L_4}{L_3}), -\arctan(\frac{L_3}{L_4})]^T$.



Configuration q_3 : Singularity-free (3 DOFs)



Configuration q_1 : Singular (2 DOFs)



Configuration q_2 : Singular (1 DOF)

Figure 4: Manipulability ellipsoids for different joint configurations as required in Exercise 7. The top figure shows a full 3D ellipsoid for a singularity-free pose. The bottom figures show the degenerated ellipsoids (a flat plane and a line) when the robot loses one or two degrees of freedom, respectively.

Exercise 8

- (a) Figure 5 replicates the kinematic chain and reference frames for the KUKA IIWA 7 robot, as provided in the lab manual.

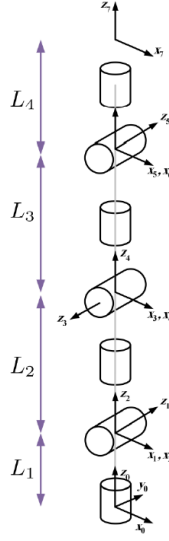


Figure 5: Illustration of the drawn reference frames for the KUKA manipulator.

- (b) Following the assigned frames and the Denavit-Hartenberg (DH) convention, the resulting DH parameters for the KUKA IIWA 7 robot are summarized in Table 2.

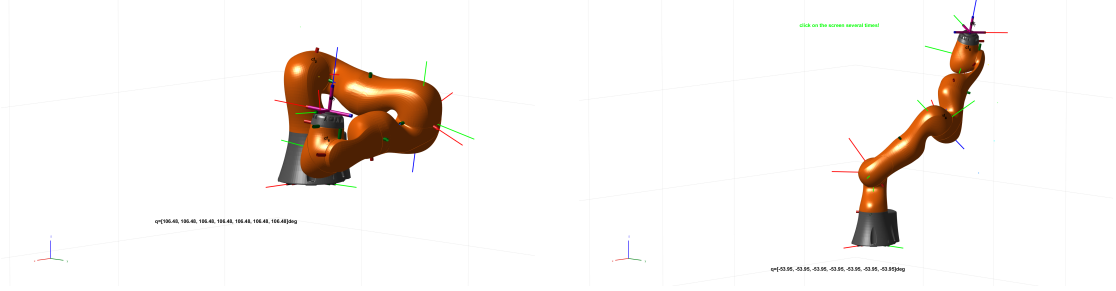
Link	θ_i	d_i	α_i	a_i
1	q_1	L_1	$-\pi/2$	0
2	q_2	0	$\pi/2$	0
3	q_3	L_2	$\pi/2$	0
4	q_4	0	$-\pi/2$	0
5	q_5	L_3	$-\pi/2$	0
6	q_6	0	$\pi/2$	0
7	q_7	L_4	0	0

Table 2: DH parameters for KUKA IIWA 7

Exercise 10

- (a) As demonstrated in Figure 6, the generated coordinate frames perfectly align with the robot's physical joints across arbitrary configurations. This visual alignment confirms the accuracy of the derived Denavit-Hartenberg parameters and the resulting absolute Homogeneous Transformation matrices for the

redundant manipulator.



(a) KUKA IIWA 7 robot model pose 1

(b) KUKA IIWA 7 robot model pose 2

Figure 6: Visual validation of the kinematic frames under arbitrary joint configurations.

Exercise 12

- (a) Similar to the 4-DOF manipulator, the analytical Jacobian matrices derived for the 7-DOF KUKA IIWA 7 were successfully validated. The error between the analytical and numerical velocities was on the order of 10^{-3} , falling well within acceptable limits and confirming the model's accuracy.

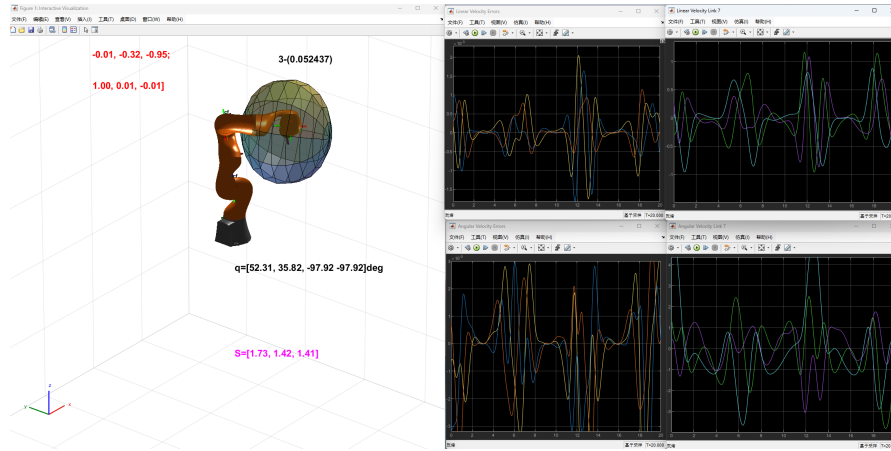


Figure 7: Manipulability ellipsoid for the KUKA robot at the specified joint configuration.

Exercise 13

- (a) To obtain the manipulability ellipsoid for the angular velocities, the Singular Value Decomposition (SVD) code must be modified to evaluate the angular

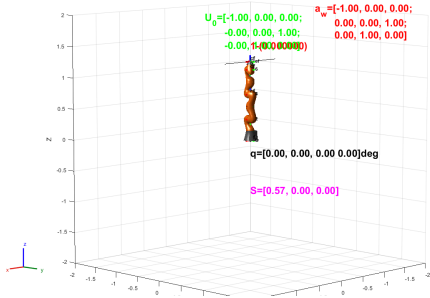
velocity Jacobian matrix, J_ω (`jef_w`), rather than the linear velocity Jacobian, J_v (`jef_v`). Specifically, the following line of code is modified:

From: `[U,S,V] = svd(jef_v,0);`

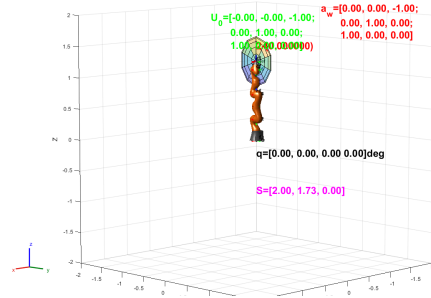
To: `[U,S,V] = svd(jef_w,0);`

The side-by-side comparison of the linear and angular manipulability ellipsoids at the zero joint configuration, $q = [0, 0, 0, 0, 0, 0]^T$, is shown in Figure 8.

As observed, at this completely stretched, vertical posture, the linear velocity ellipsoid degenerates into a 1D line (1 DOF), whereas the angular velocity ellipsoid degenerates into a 2D flat disk (2 DOFs). This phenomenon is caused by the alternating orthogonal joint axes of the KUKA robot. At $q = 0$, the rotation axes of joints 1, 3, 5, and 7 align with the global Z-axis, while joints 2, 4, and 6 align with the global Y-axis. Since no joint can provide a rotation around the global X-axis, the angular Jacobian J_ω spans only a 2D space (rank 2). Conversely, for linear velocity, the Z-axis joints (1, 3, 5, 7) contribute zero translational motion (as their rotation axes are parallel to the position vector), and the Y-axis joints (2, 4, 6) only generate instantaneous translation along the X-axis, reducing the linear Jacobian J_v to a 1D space (rank 1).



(a) Linear velocity ellipsoid (J_v , 1 DOF)



(b) Angular velocity ellipsoid (J_ω , 2 DOFs)

Figure 8: Side-by-side comparison of the linear and angular manipulability ellipsoids for the KUKA IIWA 7 at $q = [0, 0, 0, 0, 0, 0]^T$.